

Computational Astrophysics Exercises — Assignments Set 5

Numerical treatment of ordinary differential equations

Exercise 1: Local Truncation Error of Heun's Method

Consider the initial value problem $y' = f(t, y)$, $y(t_0) = y_0$. Heun's method advances the solution by one step of size h as follows

$$k_1 = f(t_n, y_n), \quad (1)$$

$$k_2 = f(t_n + h, y_n + h k_1), \quad (2)$$

$$y_{n+1} = y_n + \frac{h}{2}(k_1 + k_2). \quad (3)$$

- (a) Expand the exact solution $y(t_n + h)$ in a Taylor series about t_n up to and including terms of order h^2 . Express $y''(t_n)$ in terms of f and its partial derivatives $f_t = \frac{\partial f}{\partial t}$, $f_y = \frac{\partial f}{\partial y}$ (both evaluated at (t_n, y_n)), using the fact that $y' = f(t, y)$.
- (b) Expand $k_2 = f(t_n + h, y_n + h k_1)$ using the two-dimensional Taylor expansion of f about the point (t_n, y_n) , retaining terms up to first order in h .
- (c) Combine the results of (a) and (b) to show that the local truncation error

$$d_n = y(t_n + h) - y_{n+1} \quad (4)$$

satisfies $d_n = O(h^3)$. What is the global order of convergence of Heun's method?

(3 points)

Exercise 2: The Lane-Emden Equation

The Lane-Emden equation describes the internal structure of a (polytropic) star¹ and reads

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{dw}{d\xi} \right) + w^n = 0, \quad (5)$$

where w denotes the dimensionless density

$$\varrho = \varrho_c w^n, \quad (6)$$

with the central density ϱ_c in the star's center, and the polytropic index

$$n = \frac{1}{\gamma - 1}. \quad (7)$$

The coordinate ξ denotes the dimensionless radius (r is the radial coordinate)

$$r = \alpha \xi \quad \text{with} \quad \alpha^2 = \frac{(n+1)K}{4\pi G} \varrho_c^{\frac{1}{n}-1}. \quad (8)$$

¹We will derive the Lane-Emden equation in the exercise class.

The boundary conditions for the dimensionless Lane-Emden Equation at the center of the star (at $\xi = 0$) are

$$w = 1 \quad \text{and} \quad \frac{dw}{d\xi} = 0 \quad \text{at} \quad \xi = 0. \quad (9)$$

Analytic solutions to the Lane-Emden equation are known for the following three cases

$$n = 0 : \quad w(\xi) = 1 - \frac{1}{6}\xi^2$$

$$n = 1 : \quad w(\xi) = \frac{\sin \xi}{\xi}$$

$$n = 5 : \quad w(\xi) = \frac{1}{(1 + \xi^2/3)^{1/2}}.$$

The first case for $n = 0$ corresponds to a sphere of constant density and the case in which $n = 1$ is of some relevance in modelling planetary interiors and brown dwarfs. However, $n = 5$ leads to a model with infinite radius and finite mass.

Apart from these analytic solutions, the Lane-Emden equation has to be solved numerically, which is our goal here.

In order to integrate the Lane-Emden equation numerically, it is firstly transformed into a coupled system of first order differential equations for w and the derivative $z = \frac{dw}{d\xi}$. Then, the Lane-Emden equation is equivalent to an initial value problem for the dimensionless density w^n and its derivative z .

$$\frac{dw}{d\xi} = z \quad \text{and} \quad \frac{dz}{d\xi} = -\left(\frac{2}{\xi}z + w^n\right). \quad (10)$$

The boundary and initial conditions at the center of the star at $\xi = 0$ read

$$w = 1 \quad \text{and} \quad z = 0. \quad (11)$$

A possible and straight forward numerical scheme to solve a system of ordinary differential equations is the Heun-method.

The structure of the star is integrated from the center at $r = 0$ respecting the boundary conditions (11) with step size $\Delta\xi$.

Exercises

1. Numerical solution to the Lane-Emden equation.

- Develop a code which solves the Lane-Emden equation (5) for different n numerically using Heun's method. The first root of the function w denotes the radius of the star.
- Test your code using the analytical solutions. Plot your numerical solutions together with the analytical solutions. If you like, try different spatial accuracies by varying $\Delta\xi$.

2. Use your newly developed code for the following tasks

- a) Calculate the structure of stars for the three different values for the adiabatic exponent $\gamma = 3, 5/3$ und $7/5$, and plot the density. Specify the total mass M_* (in dimensionless units) and the radius (root ξ_n) for each case.
- b) Now, let's model the Sun: Calculate the structure of a star with one solar mass and the radius of the Sun with $n = 3$. Provide an estimation for the temperature in the center of the Sun using the ideal gas equation and parameters for the Sun. To do this, you need the following parameters: Mass of the Sun $M_\odot = 1.989 \times 10^{33}$ g, radius of the Sun $R_\odot = 6.96 \times 10^{10}$ cm, mean density of the Sun $\bar{\rho}_\odot = 1.41$ g/cm³, mean molecular weight $\mu = 0.62$. Additionally, you need to know the relation between the central density and the average density

$$\frac{\rho_c}{\bar{\rho}} = \frac{\xi_1^3}{3 \left[-\xi^2 \frac{dw}{d\xi} \right]_{\xi=\xi_1}}. \quad (12)$$

Hint: Calculate the central density ρ_c using the above relation and the average density $\bar{\rho}_\odot$. Then, calculate α using ξ_1 and $R_\odot$ and obtain K and p_c . Afterwards, use the ideal gas equation $p_c = \frac{\rho_c k_B T_c}{\mu m_H}$ to estimate the temperature in the center of the Sun. Compare your result with the number from the literature, which is approximately $T_c \approx 15.7 \times 10^6$ K.

(7 points)

A note on the use of AI:

The use of AI tools (e.g. ChatGPT, Claude, Copilot) is permitted for these exercises, provided that (1) any use is clearly documented in your submission (which tool, for what purpose), and (2) you are able to explain all submitted solutions in full — on request, during the exercise class or otherwise. Submissions that cannot be explained by the submitter(s) (that includes all members of the submitting team) will be graded as failed, regardless of whether AI was used.

in total

(A)₁₆ points

Deadline:

Fri, 12 June at noon  via email (including code) to christoph.schaefer@uni-tuebingen.de.